

The vanilla hard-wall hydrodynamic Drude model

A derivation with physical interpretation

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Abstract

This note derives the standard linearized hard-wall hydrodynamic Drude model (HDM) for a metal and explains what each equation means physically. The central modification of local Drude theory is the finite compressibility of the conduction-electron fluid. It produces a pressure term proportional to $\nabla(\nabla \cdot \mathbf{J})$, leaves the transverse response local, and gives the longitudinal electron-density response a finite wave number. The note also distinguishes the incident electromagnetic field from the total self-consistent field, explains why local Drude theory already supports surface plasmons, derives the electric-field-only formulation by eliminating $\mathbf{J}$, and derives the additional hard-wall boundary condition $\hat{\mathbf{n}} \cdot \mathbf{J} = 0$.

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1 Scope, notation, and the central idea

We consider a metal occupying a region Ω_m , surrounded by a local dielectric with relative permittivity ε_d . The unit normal $\hat{\mathbf{n}}$ points from the metal into the dielectric. The metal contains a mobile conduction-electron fluid. Bound electrons and interband processes are represented by a local background permittivity ε_∞ . The mobile electrons are treated hydrodynamically.

Throughout the derivation we use the harmonic convention

$$\text{all fields}(\mathbf{r}, t) = \Re \left[\text{field}(\mathbf{r}, \omega) e^{-i\omega t} \right]. \quad (1)$$

With this convention,

$$\partial_t \longrightarrow -i\omega. \quad (2)$$

The most important conceptual point is the following:

The HDM equations inside the metal act on the total self-consistent field $\mathbf{E}$.

(3)

For a scattering problem one may write

$$\mathbf{E}_{\text{tot}} = \mathbf{E}_{\text{inc}} + \mathbf{E}_{\text{sc}}, \quad (4)$$

but $\mathbf{E}_{\text{inc}}$ is only the prescribed external drive. The induced field $\mathbf{E}_{\text{sc}}$, including the field created by the induced charge, is part of the field that drives the electron fluid.

The local Drude model assumes

$$\mathbf{J} = \sigma_D(\omega) \mathbf{E}, \quad \sigma_D(\omega) = \frac{i\epsilon_0 \omega_p^2}{\omega + i\gamma}, \quad (5)$$

where γ is a phenomenological momentum-relaxation rate and ω_p is the plasma frequency. It contains no spatial derivative of the electron density and therefore no intrinsic microscopic length scale.

The hydrodynamic model adds the pressure force of a compressible electron fluid. In the linearized frequency-domain form its central equation is

$$\boxed{\beta^2 \nabla(\nabla \cdot \mathbf{J}) + \omega(\omega + i\gamma)\mathbf{J} = i\omega\epsilon_0\omega_p^2 \mathbf{E}.} \quad (6)$$

The first term is the nonlocal correction. It acts only on the longitudinal part of the current, because it contains $\nabla \cdot \mathbf{J}$.

The adjective hard-wall refers to the additional condition

$$\boxed{\hat{\mathbf{n}} \cdot \mathbf{J} = 0 \quad \text{at the metal boundary.}} \quad (7)$$

It says that conduction electrons cannot flow through the geometrical surface. It does *not* say that $\hat{\mathbf{n}} \cdot \mathbf{E} = 0$.

2 Why introduce a hydrodynamic model?

2.1 What local Drude theory can already do

Local Drude theory can describe collective charge oscillations at an interface. For example, a planar metal–dielectric interface supports a surface plasmon whose charge is represented, in the local idealization, by an oscillating surface sheet. A nanoparticle can similarly support a dipolar plasmon, with positive and negative induced charge patches on opposite sides.

The restoring force of these modes is the Coulomb field created by the charge separation. Fermi pressure is not required for the existence of the surface plasmon.

2.2 What is missing from local Drude theory

Local Drude theory has two related limitations:

1. the induced charge is compressed into an ideal mathematical sheet at an interface; and
2. the formal bulk condition $\epsilon_D(\omega) = 0$ gives a plasma frequency but no finite wave number or spatial structure for the bulk longitudinal oscillation.

The hydrodynamic pressure term supplies the missing spatial physics. A density gradient costs energy, so charge cannot be compressed arbitrarily sharply. The induced surface charge becomes a narrow but finite-width layer, and bulk longitudinal plasmons acquire the dispersion

$$\omega(\omega + i\gamma) = \frac{\omega_p^2}{\epsilon_\infty} + \beta^2 k^2. \quad (8)$$

Vanilla HDM is therefore a mesoscopic extension of local Drude theory: it retains the classical electromagnetic field and adds one coarse-grained quantum ingredient, namely the compressibility of the conduction-electron gas. It does not include equilibrium-density spill-out, tunnelling, or a fully self-consistent exchange–correlation functional.

3 Starting point: a charged compressible electron fluid

3.1 Fluid variables

Let $n(\mathbf{r}, t)$ be the conduction-electron number density and $\mathbf{v}(\mathbf{r}, t)$ their velocity. The electron charge is $-e$, where $e > 0$. The free-electron charge and current densities are

$$\rho_f = -en, \quad \mathbf{J} = -en\mathbf{v}. \quad (9)$$

The equilibrium jellium density is taken to be constant inside the metal, $n = n_0$, and zero outside. In the vanilla hard-wall model this sharp geometrical cutoff is imposed as an idealization:

$$n_0(\mathbf{r}) = \begin{cases} n_0, & \mathbf{r} \in \Omega_m, \\ 0, & \mathbf{r} \notin \Omega_m. \end{cases} \quad (10)$$

There is no equilibrium spill-out in this model.

3.2 Continuity equation

Conservation of the number of conduction electrons gives

$$\partial_t n + \nabla \cdot (n\mathbf{v}) = 0. \quad (11)$$

Using $\mathbf{J} = -en\mathbf{v}$, the same equation becomes charge continuity:

$$\partial_t \rho_f + \nabla \cdot \mathbf{J} = 0. \quad (12)$$

For harmonic fields,

$$-i\omega \rho_f + \nabla \cdot \mathbf{J} = 0, \quad \rho_f = \frac{1}{i\omega} \nabla \cdot \mathbf{J}. \quad (13)$$

This equation is the first reason the divergence of the current is central: charge is created wherever the current converges or terminates. A current may be large without creating charge if it is divergence-free.

3.3 Momentum equation and pressure force

The damped Euler equation for the electron fluid is

$$mn(\partial_t + \mathbf{v} \cdot \nabla) \mathbf{v} = -en\mathbf{E} - m\gamma n\mathbf{v} - \nabla p(n), \quad (14)$$

where $p(n)$ is the thermodynamic pressure of the electron gas. The three terms on the right are, respectively:

- the electric force on the negatively charged electrons;
- phenomenological momentum relaxation; and
- the pressure-gradient force, which opposes compression.

We linearize around a uniform, stationary equilibrium:

$$n = n_0 + n_1, \quad \mathbf{v} = \mathbf{v}_1, \quad |n_1| \ll n_0. \quad (15)$$

The convective term $\mathbf{v} \cdot \nabla \mathbf{v}$ is second order and is dropped. The pressure is expanded as

$$p(n_0 + n_1) = p(n_0) + \left(\frac{\partial p}{\partial n} \right)_0 n_1 + O(n_1^2). \quad (16)$$

Thus the linearized equations are

$$\partial_t n_1 + n_0 \nabla \cdot \mathbf{v}_1 = 0, \quad (17)$$

$$\partial_t \mathbf{v}_1 + \gamma \mathbf{v}_1 = -\frac{e}{m} \mathbf{E} - \frac{1}{mn_0} \left(\frac{\partial p}{\partial n} \right)_0 \nabla n_1. \quad (18)$$

Define the hydrodynamic velocity parameter β by

$$\boxed{\beta^2 \equiv \frac{1}{m} \left(\frac{\partial p}{\partial n} \right)_0} \quad (19)$$

Then the pressure force per electron mass is

$$-\frac{\beta^2}{n_0} \nabla n_1. \quad (20)$$

Physically, β is not the drift velocity of the electron gas. It is the velocity scale associated with the restoring force generated by a density gradient. It plays the role of a sound velocity for the electron fluid, but the relevant sound is a collective electronic density oscillation rather than an ordinary acoustic wave.

4 Deriving the hydrodynamic current equation

4.1 From velocity to current

Inside the uniform metal, the linearized current is

$$\mathbf{J} = -en_0 \mathbf{v}_1. \quad (21)$$

Multiply Eq. (18) by $-en_0$. The left-hand side becomes

$$-en_0(\partial_t + \gamma) \mathbf{v}_1 = (\partial_t + \gamma) \mathbf{J}. \quad (22)$$

The electric force becomes

$$(-en_0) \left(-\frac{e}{m} \mathbf{E} \right) = \frac{e^2 n_0}{m} \mathbf{E}. \quad (23)$$

Introduce the plasma frequency

$$\omega_p^2 \equiv \frac{e^2 n_0}{m \epsilon_0}. \quad (24)$$

Then the electric term is $\epsilon_0 \omega_p^2 \mathbf{E}$.

For the pressure term, use $\mathbf{J} = -en_0 \mathbf{v}_1$ in the linearized continuity equation:

$$\partial_t n_1 + n_0 \nabla \cdot \mathbf{v}_1 = 0, \quad (25)$$

$$\partial_t n_1 - \frac{1}{e} \nabla \cdot \mathbf{J} = 0. \quad (26)$$

Therefore,

$$\partial_t n_1 = \frac{1}{e} \nabla \cdot \mathbf{J}. \quad (27)$$

The pressure contribution after multiplication by $-en_0$ is

$$(-en_0) \left(-\frac{\beta^2}{n_0} \nabla n_1 \right) = e\beta^2 \nabla n_1. \quad (28)$$

In the frequency domain, Eq. (26) gives

$$-i\omega n_1 = \frac{1}{e} \nabla \cdot \mathbf{J}, \quad n_1 = \frac{i}{e\omega} \nabla \cdot \mathbf{J}. \quad (29)$$

Consequently,

$$e\beta^2 \nabla n_1 = \frac{i\beta^2}{\omega} \nabla (\nabla \cdot \mathbf{J}). \quad (30)$$

The frequency-domain current equation is therefore

$$(\gamma - i\omega)\mathbf{J} = \epsilon_0\omega_p^2\mathbf{E} + \frac{i\beta^2}{\omega}\nabla(\nabla\cdot\mathbf{J}). \quad (31)$$

Multiplying by $i\omega$ gives

$$\boxed{\beta^2\nabla(\nabla\cdot\mathbf{J}) + \omega(\omega + i\gamma)\mathbf{J} = i\omega\epsilon_0\omega_p^2\mathbf{E}.} \quad (32)$$

The sign of the pressure term is worth interpreting. In Fourier space, $\nabla \rightarrow i\mathbf{k}$, so

$$\nabla(\nabla\cdot\mathbf{J}) \longrightarrow -\mathbf{k}(\mathbf{k}\cdot\mathbf{J}). \quad (33)$$

For a longitudinal current, this is $-k^2\mathbf{J}_L$. The pressure term thus opposes short-wavelength compression and changes the longitudinal resonance condition by a positive term β^2k^2 in the dispersion relation.

4.2 Local Drude limit and polarization form

If $\beta = 0$, Eq. (32) becomes

$$\mathbf{J} = \frac{i\epsilon_0\omega_p^2}{\omega + i\gamma}\mathbf{E} \equiv \sigma_D(\omega)\mathbf{E}, \quad (34)$$

which is precisely the local Drude current.

The free-electron polarization is defined by

$$\mathbf{J} = \partial_t\mathbf{P}_f = -i\omega\mathbf{P}_f, \quad \mathbf{P}_f = \frac{\mathbf{J}}{-i\omega}. \quad (35)$$

The total displacement inside the metal is split as

$$\mathbf{D}_m = \epsilon_0\epsilon_\infty\mathbf{E} + \mathbf{P}_f. \quad (36)$$

The first term is the local background response; the second is the response of the hydrodynamic conduction electrons.

In the local limit, substituting Eq. (34) into Eq. (35) gives

$$\mathbf{P}_f = -\epsilon_0\frac{\omega_p^2}{\omega(\omega + i\gamma)}\mathbf{E}. \quad (37)$$

Therefore the familiar local Drude permittivity is

$$\boxed{\epsilon_D(\omega) = \epsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}.} \quad (38)$$

5 What should β be?

The hydrodynamic equation itself does not determine β . It is a long-wavelength closure of a microscopic electron response, so its value depends on which limit of the electron gas one is trying to reproduce.

5.1 Static compressibility: $\beta^2 = v_F^2/3$

For a three-dimensional ideal Fermi gas at zero temperature,

$$n = \frac{k_F^3}{3\pi^2}, \quad v_F = \frac{\hbar k_F}{m}, \quad E_F = \frac{1}{2}mv_F^2. \quad (39)$$

Its pressure is

$$p = \frac{2}{5}nE_F. \quad (40)$$

Since $E_F \propto n^{2/3}$, differentiation gives

$$\frac{\partial p}{\partial n} = \frac{2}{3}E_F = \frac{1}{3}mv_F^2, \quad (41)$$

$$\beta_{\text{static}}^2 = \frac{1}{m} \frac{\partial p}{\partial n} = \frac{v_F^2}{3}. \quad (42)$$

This is the equilibrium compressibility: it asks how much pressure is produced by a slowly varying change in density after the electron gas has time to re-equilibrate locally.

5.2 Collisionless optical response: $\beta^2 = 3v_F^2/5$

At optical frequencies, the perturbation can be collisionless on the time scale of the oscillation. The electron distribution then responds through Fermi-surface kinematics rather than through static thermodynamic compression. The long-wavelength expansion of the collisionless Lindhard response has the form

$$\varepsilon_L(k, \omega) - 1 \simeq -\frac{\omega_p^2}{\omega^2} \left[1 + \frac{3}{5} \frac{k^2 v_F^2}{\omega^2} + \dots \right]. \quad (43)$$

The corresponding bulk-plasmon dispersion is

$$\omega^2 = \omega_p^2 + \frac{3}{5}v_F^2 k^2 + \dots \quad (\varepsilon_\infty = 1, \gamma = 0). \quad (44)$$

Matching the hydrodynamic dispersion to this expansion gives

$$\boxed{\beta_{\text{collisionless}}^2 = \frac{3}{5}v_F^2.} \quad (45)$$

The factors $1/3$ and $3/5$ are therefore not contradictory estimates of one universal number. They describe different limits:

$$\frac{1}{3}v_F^2 \quad \text{comes from static compressibility,} \quad \frac{3}{5}v_F^2 \quad \text{from collisionless dynamic response.} \quad (46)$$

For optical plasmonics, the conventional vanilla HDM choice is often $\beta^2 = 3v_F^2/5$, because the relevant response is usually closer to the collisionless regime. The static value is also used when the model is meant to represent a quasi-static compressibility.

5.3 A frequency-dependent interpolation

A more refined long-wavelength matching to a conserving relaxation-time response gives the Halevi interpolation [3, 4]. With the present convention and a relaxation rate γ , it can be written as

$$\boxed{\beta^2(\omega) = v_F^2 \frac{\frac{3}{5}\omega + \frac{i\gamma}{3}}{\omega + i\gamma}.} \quad (47)$$

Indeed,

$$\omega \gg \gamma: \quad \beta^2(\omega) \longrightarrow \frac{3}{5}v_F^2, \quad (48)$$

$$\omega \ll \gamma: \quad \beta^2(\omega) \longrightarrow \frac{1}{3}v_F^2. \quad (49)$$

The interpolation is not an arbitrary linear blend. It is obtained by matching the order- k^2 expansion of the hydrodynamic response to a kinetic relaxation-time response with the Mermin correction, which preserves local particle-number conservation. At intermediate ω/γ , β^2 is complex. This means that the pressure-like part of the response is not exactly in phase with the density; collisions produce a phase lag.

In practice there are two separate approximations here:

1. a long-wavelength expansion in k ; and
2. a frequency interpolation in ω/γ .

The standard vanilla implementation usually takes β to be a real constant, most commonly $\sqrt{3/5}v_F$. Using Eq. (47) makes the hydrodynamic coefficient frequency-dependent and possibly complex, which is allowed but is a more refined constitutive model.

6 Transverse and longitudinal response in a homogeneous metal

6.1 The decomposition is relative to a spatial wavevector

Consider a homogeneous region and a Fourier component with wavevector $\mathbf{k}$. Any vector field can be decomposed as

$$\mathbf{E} = \mathbf{E}_T + \mathbf{E}_L, \quad \mathbf{J} = \mathbf{J}_T + \mathbf{J}_L, \quad (50)$$

where

$$\mathbf{k} \cdot \mathbf{E}_T = 0, \quad \mathbf{k} \times \mathbf{E}_L = 0, \quad \mathbf{k} \cdot \mathbf{J}_T = 0, \quad \mathbf{k} \times \mathbf{J}_L = 0. \quad (51)$$

Thus longitudinal means parallel to the relevant spatial wavevector. It does not mean parallel to the propagation direction of an externally incident electromagnetic wave unless those directions happen to coincide.

In a finite nanoparticle there need not be one unique wavevector. The transverse/longitudinal language then refers to the Helmholtz decomposition or to the modal Fourier components of the fields. For a planar interface, the in-plane wavevector is conserved, but the transverse and longitudinal modes usually have different normal wavevectors.

6.2 Transverse current: unchanged Drude response

In Fourier space,

$$\nabla(\nabla \cdot \mathbf{J}) \longrightarrow -\mathbf{k}(\mathbf{k} \cdot \mathbf{J}). \quad (52)$$

For the transverse part, $\mathbf{k} \cdot \mathbf{J}_T = 0$, so the pressure term vanishes. Equation (32) becomes

$$\mathbf{J}_T = \frac{i\epsilon_0\omega_p^2}{\omega + i\gamma} \mathbf{E}_T. \quad (53)$$

Therefore

$$\boxed{\varepsilon_T(\omega) = \varepsilon_D(\omega)}. \quad (54)$$

The hydrodynamic correction does not modify a divergence-free transverse current in a homogeneous bulk.

6.3 Longitudinal current: nonlocal response

For the longitudinal part, $\nabla(\nabla \cdot \mathbf{J}_L) \rightarrow -k^2 \mathbf{J}_L$. The current equation gives

$$\left[\omega(\omega + i\gamma) - \beta^2 k^2 \right] \mathbf{J}_L = i\omega\epsilon_0\omega_p^2 \mathbf{E}_L. \quad (55)$$

Hence

$$\mathbf{J}_L = \frac{i\omega\epsilon_0\omega_p^2}{\omega(\omega + i\gamma) - \beta^2 k^2} \mathbf{E}_L. \quad (56)$$

Using $\mathbf{D} = \epsilon_0\epsilon_\infty \mathbf{E} + \mathbf{P}_f$ and $\mathbf{P}_f = \mathbf{J}/(-i\omega)$, the longitudinal dielectric function is

$$\boxed{\epsilon_L(k, \omega) = \epsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma) - \beta^2 k^2}.} \quad (57)$$

This is spatial dispersion: the response depends on both frequency and wavevector.

6.4 Bulk longitudinal plasmons

A longitudinal bulk mode has $\mathbf{H}_L = 0$ and, in the absence of an external charge, satisfies

$$\nabla \cdot \mathbf{D} = 0. \quad (58)$$

For a nonzero longitudinal electric field this requires $\epsilon_L(k, \omega) = 0$. Therefore

$$\boxed{\omega(\omega + i\gamma) = \frac{\omega_p^2}{\epsilon_\infty} + \beta^2 k^2.} \quad (59)$$

Without damping,

$$\omega_L(k) = \sqrt{\frac{\omega_p^2}{\epsilon_\infty} + \beta^2 k^2}. \quad (60)$$

This is the bulk plasmon branch supplied by HDM. Local Drude theory has the formal condition $\epsilon_D(\omega) = 0$, but because it contains no k^2 term it cannot determine a longitudinal wavelength or spatial profile.

7 Incident field, total field, and the origin of the longitudinal drive

It is useful to separate three ideas that are often accidentally conflated.

7.1 Incident electromagnetic wave

For a homogeneous incident plane wave,

$$\mathbf{E}_{\text{inc}}(\mathbf{r}) = \mathbf{E}_0 e^{i\mathbf{k}_{\text{inc}} \cdot \mathbf{r}}, \quad \mathbf{k}_{\text{inc}} \cdot \mathbf{E}_0 = 0. \quad (61)$$

It is transverse relative to its own propagation wavevector. In homogeneous bulk, the induced local Drude current is also transverse and therefore has $\nabla \cdot \mathbf{J} = 0$. There is no bulk charge-density wave at that same plane-wave wavevector.

At normal incidence on an infinite planar interface, take $\mathbf{k}_{\text{inc}} = k\hat{\mathbf{z}}$ and tangential electric field $\mathbf{E}_0 \parallel$ interface. Even if the tangential current varies with depth because of attenuation,

$$\nabla \cdot \mathbf{J} = \partial_z J_z + \nabla_{\parallel} \cdot \mathbf{J}_{\parallel} = 0 \quad (62)$$

for the ideal translationally invariant problem: $J_z = 0$ and the tangential current has no in-plane divergence. Energy can still be dissipated through γ ; charge compression is simply not required.

7.2 Oblique incidence and boundary launching

For oblique TM incidence at a planar interface,

$$\mathbf{k}_{\text{inc}} = (q, 0, k_{z,\text{inc}}), \quad \mathbf{E}_0 = (E_x, 0, E_z), \quad qE_x + k_{z,\text{inc}}E_z = 0. \quad (63)$$

The normal field component drives a normal current. The hard wall prevents that current from crossing the interface:

$$J_{z,T}(0) + J_{z,L}(0) = 0. \quad (64)$$

The longitudinal component is therefore generated self-consistently so that the total normal electron flux vanishes. This is the precise sense in which the boundary launches the longitudinal HDM response.

The longitudinal mode has

$$\mathbf{k}_L = (q, 0, k_{z,L}), \quad \mathbf{E}_L \parallel \mathbf{k}_L, \quad (65)$$

but $k_{z,L}$ is determined by the longitudinal dispersion Eq. (59), not by the incident electromagnetic dispersion. Only the tangential momentum q is shared by the fields due to translational invariance along the interface. Consequently,

$\mathbf{E}_0 \parallel \mathbf{k}_L$ is not a general relation.

 (66)

The longitudinal field is not a second externally incident electromagnetic wave. It is part of $\mathbf{E}_{\text{tot}}$, produced by the induced charge and required by the boundary condition.

8 Surface charge in local Drude theory and finite-width charge in HDM

8.1 Why local Drude theory has surface charge

Let a planar metal occupy $z < 0$. In local Drude theory the current can be written schematically as

$$\mathbf{J}(\mathbf{r}) = \Theta(-z)\mathbf{J}_m(\mathbf{r}), \quad (67)$$

where Θ is the step function. Its divergence is

$$\nabla \cdot \mathbf{J} = \nabla \cdot [\Theta(-z)\mathbf{J}_m] \quad (68)$$

$$= \Theta(-z)\nabla \cdot \mathbf{J}_m - J_{m,z}(0)\delta(z). \quad (69)$$

The delta function is the mathematical representation of an induced surface charge:

$$\rho_f = \rho_{\text{vol}} + \sigma_s(x, y)\delta(z), \quad \sigma_s = -\frac{J_{m,z}(0)}{i\omega}. \quad (70)$$

Thus surface charge does not mean that local Drude theory violates charge continuity. The charge appears because the current is terminated at the geometrical boundary. Local Drude has no pressure-gradient energy and no microscopic length with which to resolve the termination, so the charge is represented as an infinitely thin sheet.

8.2 Local planar surface plasmon

In the quasistatic limit, take potentials of the form

$$\phi_m = Ae^{iqx+qz}, \quad z < 0, \quad \phi_d = Ae^{iqx-qz}, \quad z > 0. \quad (71)$$

Both potentials satisfy Laplace's equation away from the interface, so there is no volume charge in either homogeneous region. Nevertheless, the fields are normal to the interface with opposite signs:

$$E_{m,z}(0) = -qA, \quad E_{d,z}(0) = +qA. \quad (72)$$

Continuity of the normal displacement gives

$$\varepsilon_D(\omega)E_{m,z} = \varepsilon_d E_{d,z}, \quad \boxed{\varepsilon_D(\omega) + \varepsilon_d = 0}. \quad (73)$$

For a lossless Drude metal,

$$\varepsilon_D(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega^2}, \quad \omega_{\text{sp}} = \frac{\omega_p}{\sqrt{\varepsilon_\infty + \varepsilon_d}}. \quad (74)$$

The surface plasmon is a self-consistent oscillation: the charge produces an electric field, the field drives electron motion, and the resulting current recreates the charge. The Coulomb restoring force is sufficient for this mode; HDM is not needed for its existence.

8.3 What HDM changes

In HDM the pressure force penalizes a sharp density gradient. The charge that was a delta-function sheet in local Drude theory becomes a narrow dynamical layer inside the metal. To see its characteristic scale, take the divergence of Eq. (32):

$$\beta^2 \nabla^2 (\nabla \cdot \mathbf{J}) + \omega(\omega + i\gamma) \nabla \cdot \mathbf{J} = i\omega\epsilon_0\omega_p^2 \nabla \cdot \mathbf{E}. \quad (75)$$

Using $\nabla \cdot \mathbf{J} = i\omega\rho_f$, we need a relation between $\nabla \cdot \mathbf{E}$ and ρ_f . In the absence of externally imposed free charge, Ampere's law implies

$$\nabla \cdot \mathbf{J} = i\omega\epsilon_0\varepsilon_\infty \nabla \cdot \mathbf{E}, \quad (76)$$

and therefore

$$\rho_f = \epsilon_0\varepsilon_\infty \nabla \cdot \mathbf{E}. \quad (77)$$

Substitution into Eq. (75) yields

$$\boxed{\left[\beta^2 \nabla^2 + \omega(\omega + i\gamma) - \frac{\omega_p^2}{\varepsilon_\infty} \right] \rho_f = 0.} \quad (78)$$

Equivalently,

$$\left(\nabla^2 + k_L^2 \right) \rho_f = 0, \quad k_L^2 = \frac{\omega(\omega + i\gamma) - \omega_p^2/\varepsilon_\infty}{\beta^2}. \quad (79)$$

For frequencies below the bulk longitudinal-plasmon frequency, k_L^2 is typically negative. Writing $k_L = i\kappa_L$, the density decays into the metal over

$$\ell_L \sim \kappa_L^{-1}, \quad \kappa_L^2 \simeq \frac{\omega_p^2/\varepsilon_\infty - \omega(\omega + i\gamma)}{\beta^2}. \quad (80)$$

The rough scale β/ω_p is often quoted, although the precise decay length depends on frequency, ε_∞ , damping, and geometry.

The distinction can be summarized as follows.

Local Drude theory	Hard-wall HDM
Surface charge represented by $\sigma_s \delta(s)$	Induced charge occupies a finite layer inside the metal
No pressure length scale	Pressure introduces the scale $\ell_L \sim \beta/\omega_p$
Surface plasmons exist through Coulomb restoring force	Surface plasmons are shifted and coupled to longitudinal response
Formal bulk-plasmon frequency but no finite- k branch	Bulk longitudinal plasmons disperse as $\omega^2 \sim \omega_p^2/\varepsilon_\infty + \beta^2 k^2$

The hard-wall charge layer is confined to the metal side of the geometrical surface. This is different from self-consistent hydrodynamic or quantum hydrodynamic models, where the equilibrium density can spill out and the induced charge can extend into the dielectric region.

9 Eliminating $\mathbf{J}$: an electric-field-only equation

Many numerical implementations solve directly for $\mathbf{E}$. We now derive the corresponding equation carefully, because the term $\nabla(\nabla \cdot \mathbf{E})$ comes specifically from eliminating $\mathbf{J}$ with Maxwell's equations.

9.1 Maxwell equations inside the metal

With the convention (1), Maxwell's equations are

$$\nabla \times \mathbf{E} = i\omega\mu_0 \mathbf{H}, \quad (81)$$

$$\nabla \times \mathbf{H} = \mathbf{J} - i\omega\epsilon_0\varepsilon_\infty \mathbf{E}. \quad (82)$$

The second equation contains the conduction current separately because the background polarization has already been included in ε_∞ .

Use Faraday's law to write

$$\mathbf{H} = \frac{1}{i\omega\mu_0} \nabla \times \mathbf{E}. \quad (83)$$

Substitute this into Ampere's law:

$$\frac{1}{i\omega\mu_0} \nabla \times \nabla \times \mathbf{E} = \mathbf{J} - i\omega\epsilon_0\varepsilon_\infty \mathbf{E}, \quad (84)$$

$$\mathbf{J} = \frac{1}{i\omega\mu_0} \left[\nabla \times \nabla \times \mathbf{E} - \varepsilon_\infty k_0^2 \mathbf{E} \right], \quad k_0 = \frac{\omega}{c}. \quad (85)$$

This is the exact Maxwell relation used to remove $\mathbf{J}$.

Now take the divergence. Since the divergence of any curl is zero,

$$\nabla \cdot \mathbf{J} = \frac{1}{i\omega\mu_0} \left[\underbrace{\nabla \cdot (\nabla \times \nabla \times \mathbf{E})}_0 - \varepsilon_\infty k_0^2 \nabla \cdot \mathbf{E} \right] \quad (86)$$

$$= -\frac{\varepsilon_\infty k_0^2}{i\omega\mu_0} \nabla \cdot \mathbf{E} = i\omega\epsilon_0\varepsilon_\infty \nabla \cdot \mathbf{E}. \quad (87)$$

The last equality uses $k_0^2 = \omega^2\mu_0\epsilon_0$.

9.2 Substitution into the hydrodynamic equation

Divide the current equation (32) by $\omega(\omega + i\gamma)$. Defining

$$a \equiv \frac{\beta^2}{\omega(\omega + i\gamma)}, \quad (88)$$

we have

$$a\nabla(\nabla \cdot \mathbf{J}) + \mathbf{J} = \sigma_D \mathbf{E}. \quad (89)$$

Using Eq. (87), the current can equivalently be written

$$\mathbf{J} = \sigma_D \mathbf{E} - a\nabla(\nabla \cdot \mathbf{J}) \quad (90)$$

$$= \sigma_D \mathbf{E} - \frac{i\epsilon_0 \epsilon_\infty \beta^2}{\omega + i\gamma} \nabla(\nabla \cdot \mathbf{E}). \quad (91)$$

This equation already shows that the correction is purely longitudinal: if $\nabla \cdot \mathbf{E} = 0$, it reduces to local Drude response.

Substitute Eq. (85) and Eq. (87) directly into Eq. (89):

$$\frac{1}{i\omega\mu_0} [\nabla \times \nabla \times \mathbf{E} - \epsilon_\infty k_0^2 \mathbf{E}] - a \frac{\epsilon_\infty k_0^2}{i\omega\mu_0} \nabla(\nabla \cdot \mathbf{E}) = \sigma_D \mathbf{E}. \quad (92)$$

Multiplying by $i\omega\mu_0$ gives

$$\nabla \times \nabla \times \mathbf{E} - \epsilon_\infty k_0^2 \mathbf{E} - a\epsilon_\infty k_0^2 \nabla(\nabla \cdot \mathbf{E}) = i\omega\mu_0 \sigma_D \mathbf{E}. \quad (93)$$

Now

$$i\omega\mu_0 \sigma_D = -k_0^2 \frac{\omega_p^2}{\omega(\omega + i\gamma)}. \quad (94)$$

Combining the two local terms produces ϵ_D , and we obtain

$$\boxed{\nabla \times \nabla \times \mathbf{E} - k_0^2 \epsilon_D(\omega) \mathbf{E} = \frac{\epsilon_\infty k_0^2 \beta^2}{\omega(\omega + i\gamma)} \nabla(\nabla \cdot \mathbf{E}).} \quad (95)$$

The term on the right is absent in local electrodynamics. It is precisely the electric-field representation of the pressure response. Since $\nabla \cdot \mathbf{E}_T = 0$, it acts only on $\mathbf{E}_L$. The field in this equation is the total field, not just the incident field.

10 Boundary conditions for the hard-wall model

The hydrodynamic equation is a higher-order spatial equation than local Drude theory, so Maxwell's usual interface conditions are not sufficient by themselves. The electron fluid supplies one additional boundary condition.

10.1 Maxwell boundary conditions

Assume no externally imposed singular surface current or external free surface charge. The integral Maxwell laws give

$$\hat{\mathbf{n}} \times (\mathbf{E}_d - \mathbf{E}_m) = 0, \quad (96)$$

$$\hat{\mathbf{n}} \times (\mathbf{H}_d - \mathbf{H}_m) = 0, \quad (97)$$

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_d - \mathbf{B}_m) = 0, \quad (98)$$

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_d - \mathbf{D}_m) = 0. \quad (99)$$

The last equation is written for the total displacement, including the hydrodynamic free-electron polarization inside the metal.

Outside the metal,

$$\mathbf{D}_d = \epsilon_0 \epsilon_d \mathbf{E}_d. \quad (100)$$

Inside,

$$\mathbf{D}_m = \epsilon_0 \epsilon_\infty \mathbf{E}_m + \mathbf{P}_f, \quad \mathbf{P}_f = \frac{\mathbf{J}}{-i\omega}. \quad (101)$$

10.2 Derivation of the hard-wall additional boundary condition

The hard-wall assumption is a statement about particle transport:

$$\hat{\mathbf{n}} \cdot \mathbf{J}_m = 0. \quad (102)$$

Because $\mathbf{J} = -i\omega \mathbf{P}_f$, this is equivalent for $\omega \neq 0$ to

$$\hat{\mathbf{n}} \cdot \mathbf{P}_f = 0. \quad (103)$$

Insert this into the normal-displacement condition:

$$\hat{\mathbf{n}} \cdot \mathbf{D}_d = \hat{\mathbf{n}} \cdot \mathbf{D}_m, \quad (104)$$

$$\epsilon_0 \epsilon_d E_{d,n} = \epsilon_0 \epsilon_\infty E_{m,n} + P_{f,n}, \quad (105)$$

$$\epsilon_d E_{d,n} = \epsilon_\infty E_{m,n}. \quad (106)$$

For vacuum outside, $\epsilon_d = 1$, so

$$E_{d,n} = \epsilon_\infty E_{m,n}. \quad (107)$$

The electric field inside the metal is therefore not multiplied by ϵ_∞ . Rather, the normal boundary relation contains ϵ_∞ because the normal component of the conduction-electron polarization vanishes at a hard wall. The remaining normal displacement is the background part $\epsilon_0 \epsilon_\infty \mathbf{E}_m$.

The standard hard-wall interface conditions can consequently be listed as

$$\begin{aligned} \hat{\mathbf{n}} \times (\mathbf{E}_d - \mathbf{E}_m) &= 0, \\ \hat{\mathbf{n}} \times (\mathbf{H}_d - \mathbf{H}_m) &= 0, \\ \epsilon_d E_{d,n} &= \epsilon_\infty E_{m,n}, \\ \hat{\mathbf{n}} \cdot \mathbf{J}_m &= 0. \end{aligned} \quad (108)$$

The last condition is the additional boundary condition (ABC) required by HDM. It is a no-flux condition, not a no-field condition:

$$\hat{\mathbf{n}} \cdot \mathbf{J} = 0 \quad \not\Rightarrow \quad \hat{\mathbf{n}} \cdot \mathbf{E} = 0. \quad (109)$$

10.3 Electric-field-only form of the ABC

When solving the electric-field equation (95), one can replace $\mathbf{J}$ using Eq. (85). The hard-wall condition becomes

$$\hat{\mathbf{n}} \cdot [\nabla \times \nabla \times \mathbf{E}_m - \epsilon_\infty k_0^2 \mathbf{E}_m] = 0. \quad (110)$$

The omitted prefactor $1/(i\omega\mu_0)$ is nonzero and therefore does not affect the zero condition.

10.4 Two bookkeeping conventions for surface charge

There are two equivalent ways to describe interface charge, provided one does not mix conventions.

In the total-displacement convention used above, the hydrodynamic conduction electrons are included in $\mathbf{D}_m$, so there is no externally imposed delta-function sheet in the normal $\mathbf{D}$ boundary condition. Their charge is encoded in $\nabla \cdot \mathbf{P}_f$ inside the metal.

Alternatively, one may treat $\mathbf{P}_f$ as a separate polarization and write its divergence explicitly as the conduction-electron charge. In local Drude theory the sharp step in $\mathbf{P}_f$ produces a delta-function sheet. In hard-wall HDM, $P_{f,n} = 0$ prevents such a singular normal-polarization jump at the ideal wall, while the pressure term resolves the charge into a finite-width volume layer just inside the metal.

11 What exactly is ε_∞ ?

11.1 Physical meaning

The decomposition

$$\mathbf{D}_m = \epsilon_0 \varepsilon_\infty \mathbf{E} + \mathbf{P}_f \quad (111)$$

separates two electronic responses:

1. $\mathbf{P}_f$: the explicitly modeled mobile conduction-electron fluid; and
2. $\epsilon_0(\varepsilon_\infty - 1)\mathbf{E}$: a local background polarization due to bound/core electrons and interband transitions that are not included in the hydrodynamic fluid.

For an ideal jellium gas in vacuum, there are no bound electrons in the model, so

$$\varepsilon_\infty = 1. \quad (112)$$

For a real metal, ε_∞ is generally an effective material parameter. It is not the full measured metal permittivity at the operating frequency. The Drude part must not be counted twice.

11.2 How ε_∞ is obtained

In the simplest Drude–HDM model, one fits the measured complex permittivity over a selected frequency window to

$$\varepsilon_{\text{exp}}(\omega) \simeq \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}. \quad (113)$$

Depending on what is known independently, one may fit ε_∞ , ω_p , and γ simultaneously, or fix ω_p and γ and fit only the background.

Equivalently, a provisional value can be extracted at a given frequency as

$$\varepsilon_\infty \simeq \varepsilon_{\text{exp}}(\omega) + \frac{\omega_p^2}{\omega(\omega + i\gamma)}, \quad (114)$$

although fitting over a frequency interval is more reliable.

If interband transitions are important, they should be modeled explicitly, for example with a Drude–Lorentz form:

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)} + \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\Gamma_j\omega}. \quad (115)$$

In that case ε_∞ is the residual high-frequency local background after the explicitly included Drude and Lorentz contributions have been separated. In a fully realistic material model it can itself be complex and frequency-dependent.

The practical rule is therefore:

ε_∞ is fitted or calibrated, not a universal metal constant.

(116)

12 A compact physical picture

The entire response mechanism can be read as a sequence:

1. The total electric field drives the conduction-electron current.
2. A divergence-free tangential current remains transverse and follows the local Drude response.
3. A normal current driven toward a hard wall cannot cross it, so the current must terminate or be redirected near the boundary.
4. Current convergence creates charge through $\rho_f = (i\omega)^{-1}\nabla\cdot\mathbf{J}$.
5. The charge creates a self-consistent longitudinal field $\mathbf{E}_L = -\nabla\phi_L$.
6. Coulomb forces produce the familiar surface-plasmon restoring force; finite pressure spreads the charge and adds the longitudinal $\beta^2 k^2$ dispersion.
7. The hard-wall condition enforces zero normal electron flux while allowing a nonzero normal electric field.

The two limiting pictures are therefore compatible:

local Drude: ideal surface-charge sheet $\longrightarrow$ HDM: finite-width dynamical charge layer.	(117)
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13 The vanilla hard-wall HDM in one place

For reference, the model consists of the following ingredients.

Bulk equations inside the metal

$$\nabla \times \mathbf{E} = i\omega\mu_0\mathbf{H}, \quad (118)$$

$$\nabla \times \mathbf{H} = \mathbf{J} - i\omega\epsilon_0\epsilon_\infty\mathbf{E}, \quad (119)$$

$$\beta^2\nabla(\nabla\cdot\mathbf{J}) + \omega(\omega + i\gamma)\mathbf{J} = i\omega\epsilon_0\omega_p^2\mathbf{E}. \quad (120)$$

Interface conditions

$$\hat{\mathbf{n}} \times (\mathbf{E}_d - \mathbf{E}_m) = 0, \quad (121)$$

$$\hat{\mathbf{n}} \times (\mathbf{H}_d - \mathbf{H}_m) = 0, \quad (122)$$

$$\epsilon_d E_{d,n} = \epsilon_\infty E_{m,n}, \quad (123)$$

$$\hat{\mathbf{n}} \cdot \mathbf{J}_m = 0. \quad (124)$$

Material parameters

$$\omega_p^2 = \frac{e^2 n_0}{m\epsilon_0}, \quad \sigma_D = \frac{i\epsilon_0\omega_p^2}{\omega + i\gamma}, \quad \epsilon_D = \epsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}. \quad (125)$$

The coefficient β is chosen according to the response being matched; the common collisionless optical choice is $\beta^2 = 3v_F^2/5$, while static compressibility gives $\beta^2 = v_F^2/3$.

What is and is not included

Vanilla hard-wall HDM includes finite compressibility, nonlocal longitudinal response, and a no-flux electron boundary condition. It does not include:

- equilibrium-density spill-out;
- tunnelling through a dielectric gap;
- explicit exchange–correlation potentials;
- microscopic surface scattering or Landau damping; or
- a spatially varying equilibrium density.

Those effects require extensions such as self-consistent quantum hydrodynamics, the generalized nonlocal optical response model, quantum-corrected models, or surface-response/Feibelman-parameter descriptions.

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